

Bangladesh University of Business and Technology



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Lab Report

Experiment No : 02 & 03
Course Title : Object-Oriented Programming
Course Code : CSE 111 & 112
Experiment Name : Lab Report-2 & Lab report-3.

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Program No: 01

Program Name: You are required to implement two classes, Varsity and Department. The Varsity class contains a private member 'name' and a protected member 'location'. Implement a default constructor in this class to set the value of name and location. The Department class has a member function 'Print', which takes an object of type Varsity as a parameter and is declared as a friend of the Varsity class, allowing it to access the private and protected members of Varsity. In the main function, create an object of Varsity and an object of Department. Use the Print function to output the values of the name and location of the Varsity object.

Source Code:

```
#include <iostream>

#include <string>

using namespace std;

class Varsity{

    string name;

    protected:

        string location;

    public:

        Varsity() {

            name = "BRAC University";

            location = "Bangladesh";

        }
```

```

        friend class Department;

};

class Department{

    public:

        void Print(Varsity x){

            cout << "Varsity Name :" << x.name << "Location :" << x.location<< endl;

        }

};

int main(){

    Varsity x;

    Department y;

    y.Print(x);

    return 0;

}

```

Output:

```

abdul@Al-Noman MINGW64 /e/BUBT Assignment/Lab-2 (main)
● $ g++ problemOne.cpp -o problemOne

abdul@Al-Noman MINGW64 /e/BUBT Assignment/Lab-2 (main)
● $ ./problemOne.exe
Varsity Name :BRAC UniversityLocation :Bangladesh
❖
abdul@Al-Noman MINGW64 /e/BUBT Assignment/Lab-2 (main)
○ $ 

```

Program No: 02

Program Name: You are required to implement two classes, Student and Faculty. The Student class contains a private member studentID and a protected member name. Implement a parameterized constructor to set the value of studentID and name. The Faculty class has a member function Display, which takes an object of type Student as a parameter and is declared as a friend of the Student class, allowing it to access the private and protected members of Student. In the main function, create an object of Student and an object of Faculty. Use the Display function to output the values of the studentID and name of the Student object.

Source Code:

```
#include <iostream>

#include <string>

using namespace std;

class Student{

    int studentId;

    protected:

    string name;

    public:

    Student(int studentId, string name){

        this->studentId = studentId;

        this->name = name;

    }
```

```

    friend class Faculty;

};

class Faculty{

    public:

    void Display(Student x){

        cout << "Student Id : " << x.studentId << " Name : " << x.name << endl;

    }

};

int main(){

    Student s1( 202321," MR Nothing ");

    Faculty y;

    y.Display(s1);

    return 0;

}

```

Output:

```

abdul@Al-Noman MINGW64 /e/BUBT Assignment/Lab-2 (main)
● $ g++ problemTwo.cpp -o problemTwo

abdul@Al-Noman MINGW64 /e/BUBT Assignment/Lab-2 (main)
● $ ./problemTwo.exe
Student Id : 202321 Name : MR Nothing
☐
abdul@Al-Noman MINGW64 /e/BUBT Assignment/Lab-2 (main)
○ $ ☐

```

Program No: 03

Program Name: You are required to implement a class, Point2D. The Point2D class contains private members x and y to represent the coordinates of a point. Implement a parameterized constructor to initialize these values. There is a global function calculateDistance(), which takes two Point2D objects as parameters and is declared as a friend of the Point2D class, allowing it to access the private members x and y. The calculateDistance() function calculates the distance between two points in a 2D coordinate system. In the main function, create two Point2D objects, display their coordinates, and use the calculateDistance() function to compute and output the distance between the two points.

Source Code:

```
#include <iostream>

#include <cmath>

using namespace std;

class Point2D{

    double x, y;

    public:

    Point2D( double x, double y){

        this->x = x;

        this->y = y;

    }

    void Display(){

        cout << "x = " << x << " Y = " << y << endl;

    }

    friend double calculateDistance(Point2D a, Point2D b);
```

```
};
```

```
double calculateDistance(Point2D a,Point2D b){
```

```
    double dx = a.x - b.x;
```

```
    double dy = a.y - b.y;
```

```
    return sqrt(dx * dx + dy * dy);
```

```
}
```

```
int main(){
```

```
    Point2D p1(1,2), p2(2, 5);
```

```
    p1.Display();
```

```
    p2.Display();
```

```
    cout << "Distance = " << calculateDistance(p1, p2) << endl;
```

```
    return 0;
```

```
}
```

Output:

```
abdul@Al-Noman MINGW64 /e/BUBT Assignment/Lab-2 (main)
● $ g++ problemThree.cpp -o problemThree

abdul@Al-Noman MINGW64 /e/BUBT Assignment/Lab-2 (main)
● $ ./problemThree
x = 1 Y = 2
x = 2 Y = 5
Distance = 3.16228

abdul@Al-Noman MINGW64 /e/BUBT Assignment/Lab-2 (main)
○ $
```

Program No: 04

Program Name: Define a class Person with private members for name and age. Implement a parameterized constructor to initialize name and age. Define a friend function compareAges that takes two Person objects and prints which person is older.

Source Code:

```
#include <iostream>

#include <string>

using namespace std;

class Person {

private:

    string name;

    int age;

public:

    Person(string n, int a) {

        name = n;

        age = a;

    }

    friend void compareAges(Person p1, Person p2);

};
```



```
void compareAges(Person p1, Person p2) {  
    if (p1.age > p2.age) {  
        cout << p1.name << " is older." << endl;  
    }  
    else if (p2.age > p1.age) {  
        cout << p2.name << " is older." << endl;  
    }  
    else {  
        cout << "Both are same age." << endl;  
    }  
}
```

```
int main() {  
    Person p1("Noman", 22);  
    Person p2("Maisa", 23);  
  
    compareAges(p1, p2);  
  
    return 0;  
}
```

Output:

```
abdul@Al-Noman MINGW64 /e/BUBT Assignment/Lab-2 (main)
• $ g++ ./p4.cpp -o p4

abdul@Al-Noman MINGW64 /e/BUBT Assignment/Lab-2 (main)
$ ./p4.exe
• Maisa is older.
✖
abdul@Al-Noman MINGW64 /e/BUBT Assignment/Lab-2 (main)
○ $ █
```

Lab Report -3

Program No: 05

Program Name: Store book title and price in an array of Book objects.

Source Code:

```
#include <iostream>
```

```
#include <string>
```

```
using namespace std;
```

```
class Book{
```

```
public:

    string title;

    float price;

};


int main(){

    Book b[5];

    for(int i = 0; i < 5; i++){

        cout << "Enter details for book " << i + 1 << endl;

        cout << " Title ";

        cin >> b[i].title;

        cout << " Price ";

        cin >> b[i].price;

    }

    for(int i = 0; i < 5; i++){

        cout << "Book " << i + 1 << endl;

        cout << " Title: " << b[i].title << " Price: " << b[i].price << endl;

    }
```

```

int index = 0;

for (int i = 1; i < 5; i++){
    if (b[i].price > b[index].price){
        index = i;
    }
}

cout << "Most expensive books is = " << "Title: " << b[index].title << "Price: "
<< b[index].price<< endl;

return 0;
}

```

Output:

```

abdul@Al-Noman MINGW64 /e/BUBT Assignment/Lab-2 (main)
$ g++ ./p5.cpp -o p5

abdul@Al-Noman MINGW64 /e/BUBT Assignment/Lab-2 (main)
$ ./p5.exe
Enter details for book 1
  Title AD
  Price 32
Enter details for book 2
  Title Sd
  Price 21
Enter details for book 3

```

```
BOOK 3
Title: fs Price: 123
Most expensive books is = Title: AsdasdPrice: 1234
abdul@Al-Noman MINGW64 /e/BUBT Assignment/Lab-2 (main)
$
```

Program No: 06

Program Name: Create an array of Distance objects and calculate the sum of all distances.

Source Code:

```
#include <iostream>

using namespace std;
```

```
class Distance{
```

```
public:
```

```
double value;
```

```
Distance(){
```

```
    value = 0;
```

```
}
```

```
void input(){
```

```
    cout << "Enter Distance: ";
```

```
    cin >> value;
```

```
}
```

```
void Display(){
```

```
    cout << value << endl;
```

```
}
```

```
};
```

```
int main(){
```

```
    Distance d[2];
```

```
    double sum = 0;
```

```
    for (int i = 0; i < 5; i++) {
```

```
        cout << "Distance " << i + 1 << endl;
```

```
        d[i].input();
```

```
    }
```

```
    for (int i = 0; i < 5; i++) {
```

```
        d[i].Display();
```

```
        sum += d[i].value;
```

```
    }
```

```
    cout << "\nTotal Sum = " << sum << endl;
```

```
    return 0;
```

}

Output:

```
abdul@Al-Noman MINGW64 /e/BUBT Assignment/Lab-2 (main)
● $ ./p6.exe
Distance 1
Enter Distance: 12
Distance 2
Enter Distance: 32
Distance 3
Enter Distance: 12
12
32
12
56
1.79169e-307

Total Sum = 112
abdul@Al-Noman MINGW64 /e/BUBT Assignment/Lab-2 (main)
```

Program No: 07

Program Name: Create an array of Distance objects and calculate the sum of all distances.

Source Code:

```
#include <iostream>

using namespace std;
```

```
class Rectangle{

    int width, length;
```

public:

void areaSetData(int width, int length){

 this->width = width;

 this->length = length ;

}

int CalculateArea(){

 return width * length;

}

};

int main(){

 Rectangle r[3];

 int width, length;

 for (int i = 0; i < 3; i++){

 cout << "Enter the width and length of rectangle" << i + 1 << endl;

 cin >> width >> length;

 r[i].areaSetData(width, length);

 };


```
int maxArea = r[0].CalculateArea();

int maxIndex = 0;

for (int i = 0; i < 3; i++){

    int area = r[i].CalculateArea();

    cout << "Rectangle " << i + 1 << " Area : " << area << endl;

    if(area > maxArea){

        maxArea = area;

        maxIndex = i;

    }

}

cout << "Rectangle " << maxIndex + 1 << " has the largest area." << endl;

return 0;

}
```

Output:

```
abdul@Al-Noman MINGW64 /e/BUBT Assignment/Lab-2 (main)
$ ./p7.exe
4
5
Enter the width and length of rectangle3
6
5
Rectangle 1 Area : 6
Rectangle 2 Area : 20
Rectangle 3 Area : 30
Rectangle 2 has the largest area.
abdul@Al-Noman MINGW64 /e/BUBT Assignment/Lab-2 (main)
```

Program No: 08

Program Name: Store account number and balance.

- Create an array of Account objects.
- Display all account details.
- Find accounts having balance less than 1000.

Source Code:

```
#include <iostream>
using namespace std;

class Account{
    int account_Number;
    double balance;
```

public:

```
void accountDetails(int accountNumber, double balance){  
    this->account_Number = accountNumber;  
    this->balance = balance;  
}
```

```
void showData(){  
    cout << "Account Number: " << account_Number  
        << " Balance: " << balance << endl;  
}
```

```
double getBalance(){  
    return balance;  
}  
};
```

```
int main(){  
    Account a[3];  
  
    int accNo;  
    double bal;  
  
    for(int i = 0; i < 3; i++){  
        cout << "Enter account number and balance for account " << i + 1 << endl;  
        cin >> accNo >> bal;
```

```
        a[i].accountDetails(accNo, bal);
    }

    for(int i = 0; i < 3; i++){
        if(a[i].getBalance() < 1000){
            a[i].showData();
        }
    }

    return 0;
}
```

Program No: 09

Program Name: Store product name, quantity, and price.

- Input 10 products.
- Calculate total inventory value.
- Inventory Value=Quantity×Price
- Find the product with maximum inventory value.

Source Code:

```
#include <iostream>

using namespace std;
```

```
class Product
{
    string name;
    int quantity;
    double price;

public:
    void ProductInventory(string name, int quantity, double price){
        this->name = name;
        this->quantity = quantity;
        this->price = price;
    }

    double productValue(){
        return quantity * price;
    }

    void showProduct(){
        cout << "Name: " << name << "\n"
              << "Quantity: " << quantity << "\n"
              << "Price: " << price << "\n"
              << "Product value: " << productValue() << endl;
    }
}
```

```
};
```

```
int main(){
```

```
    Product p[10];
```

```
    string name;
```

```
    int quantity;
```

```
    double price;
```

```
    double totalValue = 0;
```

```
    int maxIndex = 0;
```

```
    double maxValue = 0;
```

```
    for (int i = 0; i < 10; i++){
```

```
        cout << "Enter product name, quantity, price for product " << i + 1 << endl;
```

```
        cin >> name >> quantity >> price;
```

```
        p[i].ProductInventory(name, quantity, price);
```

```
    }
```

```
    maxValue = p[0].productValue();
```

```
    for (int i = 0; i < 10; i++){
```

```
double v = p[i].productValue();
```

```
p[i].showProduct();
```

```
totalValue += v;
```

```
if (v > maxValue){
```

```
    maxValue = v;
```

```
    maxIndex = i;
```

```
}
```

```
}
```

```
cout << "Product with Maximum Value:" << endl;
```

```
p[maxIndex].showProduct();
```

```
return 0;
```

```
}
```

Output:

```
abdul@Al-Noman MINGW64 /e/BUBT Assignment/Lab-2 (main)
```

```
$ ./product.exe
```

```
Name: tre
```

```
Quantity: 32
```

```
Price: 12312
```

```
Product value: 393984
```

```
Product with Maximum Value:
```

```
Name: y
```

```
Quantity: 122
```

```
Price: 4223
```

```
Product value: 515206
```

```
abdul@Al-Noman MINGW64 /e/BUBT Assignment/Lab-2 (main)
```

```
$
```